1. The solution of $(e^x+1)ydy=(y+1)e^x dx$ is

NIMCET 2017

(a)$e^x=c(e^x+1)(y+1)$
(b)$e^y=e^x+y+1$
(c)$y=(e^x+1)(y+1)$
(d)None of the above

2. The solution of the differential equation

$$\frac{dy}{dx}=e^{x+y}+x^2e^y \text{ is}$$

NIMCET 2017

(a)$e^{x-y}+\frac{x^3}{3}=C$
(b)$e^x+e^{-y}+\frac{x^3}{3}=C$
(c)$e^x-e^{-y}+\frac{x^3}{3}=C$
(d)None of the above

3. Through any point (x, y) of a curve which passes through the origin, lines are drawn parallel to the coordinate axes. The curve given that divides the rectangle formed by the two lines and the axes into two areas, one of the which is twice the other, represent a family of

NIMCET 2018

(a)circles (b) parabolas
(c)hyperbolas (d) straight line

4. The curve satisfying the differential equation $ydx-(x+3y^2)dy=0$ and passing through the point (1, 1) also passes through the point.......

NIMCET 2019

(a) $\left(\frac{1}{4},\frac{1}{2}\right)$ (b) $\left(\frac{1}{4},-\frac{1}{2}\right)$
(c) $\left(-\frac{1}{3},\frac{1}{3}\right)$ (d)$\left(\frac{1}{3},-\frac{1}{3}\right)$

5. Let $f(x)$ be a polynomial satisfying $f(0)=2, f'(0)=3$ and $f''(x)=f(x)$. Then, $f(4)$ is equal to

NIMCET 2019

(a)$\frac{5(e^8-1)}{2e^4}$ (b)$\frac{5e^8-1}{2e^4}$
(c)$\frac{2e^8}{5e^8-1}$ (d)$\frac{2e^4}{5(e^8+1)}$

6. If $(x)=x^n$, then $\boldsymbol{f(1)}+\frac{f'(1)}{1!}+\frac{f''(1)}{2!}+\cdots\frac{f^n(1)}{n!}$ is equal to: (Differentiation)

(a)0 (b)2^n
(c)2^{n-1} (d)$\frac{n(n+1)}{2}$

NIMCET 2024

ANSWER KEY

1.	D	2.	B	3.	B	4.	C	5.	B
6.	B								

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	00
NIMCET 2008	00	NIMCET 2017	02
NIMCET 2009	00	NIMCET 2018	01
NIMCET 2010	00	NIMCET 2019	02
NIMCET 2011	00	NIMCET 2020	00
NIMCET 2012	00	NIMCET 2021	00
NIMCET 2013	00	NIMCET 2022	00
NIMCET 2014	00	NIMCET 2023	00
NIMCET 2015	00	NIMCET 2024	01
		NIMCET 2025	00

SOLUTION

1. **(d)** We have,

$(e^x+1)ydy=(y+1)e^x dx \Rightarrow \frac{y}{y+1}dy=\frac{e^x}{e^x+1}dx$

On integrating both the sides, we get

$\int \frac{y}{y+1}dy=\int \frac{e^x}{e^x+1}dx \Rightarrow \int \left(1-\frac{1}{1+y}\right)dy=\int \frac{e^x}{e^x+1}dx$

$\Rightarrow y=\log(y+1)=\log(e^x+1)+\log C$

$\Rightarrow y=\log(y+1)+\log(e^x+1)+\log C$

$\Rightarrow y=\log[C(y+1)(e^x+1)]$

$\Rightarrow e^y=C(y+1)(e^x+1)$

2. **(b)** Given differential equation is

$\frac{dy}{dx}=e^{x+y}+x^2e^y=e^y(x^2+e^x)$

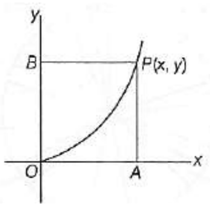
$\Rightarrow \frac{dy}{e^y}=(x^2+e^x)dx$ [Using variable separable form]

Integrating both sides, we get

$\int e^{-y}dy=\int (x^2+e^x)dx \Rightarrow -e^{-y}=\frac{x^3}{3}+e^x+C$

$\Rightarrow e^x+e^{-y}+\frac{x^3}{3}=C$ [where, C is constant]

3. **(b)** Area of OPB = 2 (area of OAP)



$\therefore xy-\int_0^x ydx=2\int_0^x ydx$

$\Rightarrow xy=3\int_0^x ydx$

Differentiation with respect to x, we get

$y+x\frac{dy}{dx}=3y \Rightarrow \frac{dy}{y}=2\frac{dx}{x}\Rightarrow \log y$

$=2\log x+\log c\Rightarrow \log y=\log cx^2\Rightarrow y=cx^2$

This is an integrating yields $y=cx^2$.

Hence, it is a parabola.

4. **(c)** We have, $ydx - (x + 3y^2)dy = 0$

$\Rightarrow ydx - xdy = 3y^2dy \Rightarrow \frac{ydx - xdy}{y^2}$

$= 3dy \Rightarrow \int d\left(\frac{x}{y}\right) = 3\int dy$

On integrating both sides, we get $\frac{x}{y} = 3y + C$

This curve is passes through (1,1)

$\therefore 1 = 3 + C$ Hence, $C = -2$

Finally curve is $\frac{x}{y} = 3y - 2$

It passes through $\left(-\frac{1}{3}, \frac{1}{3}\right)$.

5. **(b)** $f(x)$ be a polynomial satisfying.

$f(0) = 2, f'(0) = 3, f'(x) = f(x), f(4) = ?$

$f''(x) = f(x) \Rightarrow \frac{d^2y}{dx^2} = y$

Solution of this differential equation is

$y = Ae^x + Be^{-x}$

$f(x) = Ae^x + Be^{-x} \Rightarrow f(0) = A + B$

$f'(x) = Ae^x - Be^{-x} \Rightarrow f'(0) = A - B$

$A + B = 2$

$\underline{A - B = 3}$

$2A = 5$

$A = \frac{5}{2}, B = -\frac{1}{2}$

$f(x) = \frac{5}{2}e^x - \frac{1}{2}e^{-x}$

$f(4) = \frac{5}{2}e^4 - \frac{1}{2}e^{-4} = \frac{5(e^8) - 1}{2e^4}$

6. **(b)** $f(x) = x^n$ then $f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} \ldots\ldots\ldots \frac{f^n(1)}{n!}$

$f(x) = x^n$ $\quad$ $f(1) = 1$

$f'(x) = nx^{n-1}$ $\quad$ $f'(1) = n$

$f''(x) = n(n-1)x^{n-2}$ $\quad$ $f''(1) = n(n-1)$

$f^n(x) = n(n-1)(n-2) \ldots\ldots 1$ $\quad$ $f^n(1) = n!$

$f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \ldots\ldots\ldots\ldots$

$1 + \frac{n}{1!} + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \cdots \ldots \frac{n!}{n!}$

${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots {}^nC_n$

$= 2^n$